

Experiment No.	2
Experiment Title	Email Spam/Ham Classification using Naive Bayes and KNN
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1 Objective

The objective of this experiment is to classify email messages as Spam or Ham using machine learning algorithms. The SpamBase dataset is preprocessed before model training, and Gaussian Naive Bayes, Multinomial Naive Bayes, Bernoulli Naive Bayes, and K-Nearest Neighbors (KNN) classifiers are implemented for spam classification. Finally, the models are evaluated using performance metrics to determine the most effective classifier for spam detection.

2 Problem Statement

Email spam detection is a binary classification problem in machine learning. The objective of this experiment is to classify email messages as either **Spam** or **Ham (Not Spam)** using the SpamBase dataset. Different Naive Bayes classifiers and the K-Nearest Neighbors (KNN) algorithm are implemented and compared using standard performance metrics to identify the best-performing classifier.

3 Dataset Description

The SpamBase dataset is a benchmark dataset used for email spam classification. It contains numerical features extracted from email messages, such as word frequencies, character frequencies, and capital letter statistics.

Dataset Property	Description
Dataset Name	SpamBase Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	4601 email messages
Number of Features	57 input features
Target Variable	class (0 = Ham, 1 = Spam)
Number of Classes	2
Missing Values	None
Train-Test Split	80% Training, 20% Testing

Dataset Overview

The SpamBase dataset consists of numerical features representing the frequency of specific words, special characters, and capital letters in email messages. These features are used to classify emails into two categories: Spam and Ham. Since the dataset contains no missing values, only duplicate checking and feature scaling were required before model training.

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset before training the machine learning models.

4.1 Dataset Overview

The SpamBase dataset contains numerical attributes extracted from email messages. These attributes represent the frequency of commonly occurring words, special characters, and capital letter statistics that help distinguish spam emails from legitimate emails.

- Number of samples: 4601
- Number of features: 57
- Target variable: class
- Number of classes: 2 (Spam and Ham)

4.2 Statistical Summary

The statistical summary provides information about the distribution of numerical features such as mean, standard deviation, minimum, and maximum values. Since all input attributes are numerical, descriptive statistics help understand the range and variability of the dataset.

4.3 Missing Value Analysis

The dataset contains no missing values. Therefore, no missing value imputation was required before model training.

4.4 Class Distribution

The class distribution was analyzed using a count plot to determine whether the dataset contains a balanced number of spam and ham emails.

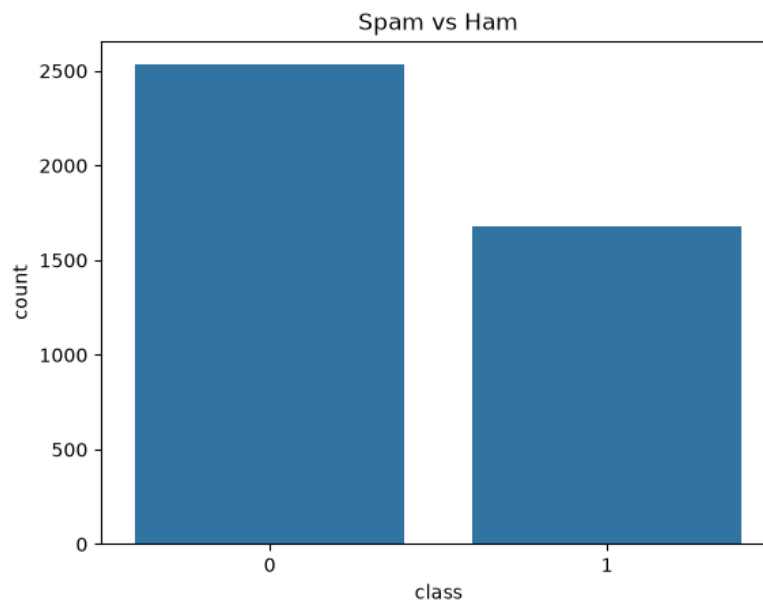


Figure 1: Distribution of Spam and Ham Emails

Inference:

- The dataset contains two classes: Spam and Ham.
- Both classes are reasonably represented in the dataset.
- The class distribution indicates that the dataset is suitable for binary classification.
- Understanding the class distribution helps in selecting appropriate evaluation metrics.

4.5 Correlation Matrix

A correlation heatmap was generated to study the relationship among numerical features in the SpamBase dataset.

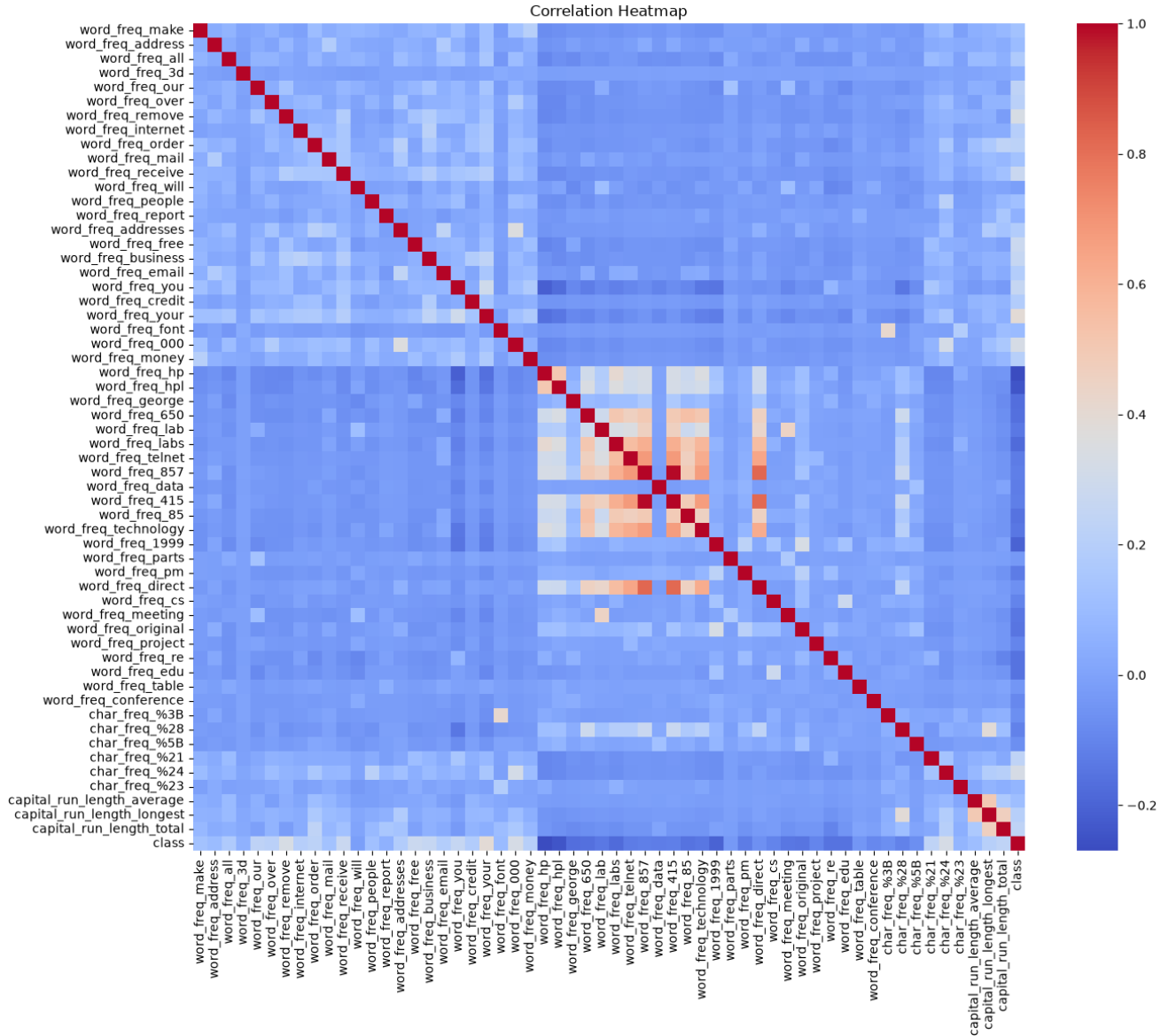


Figure 2: Correlation Heatmap of SpamBase Features

Inference:

- Most features exhibit low to moderate correlation with one another.
- No severe multicollinearity is observed among the input features.
- Some word-frequency features show positive correlation with the target class.
- The correlation matrix helps understand feature relationships before model training.

4.6 Feature Distribution

Histograms and box plots were used to visualize the distribution of feature values and identify possible outliers.

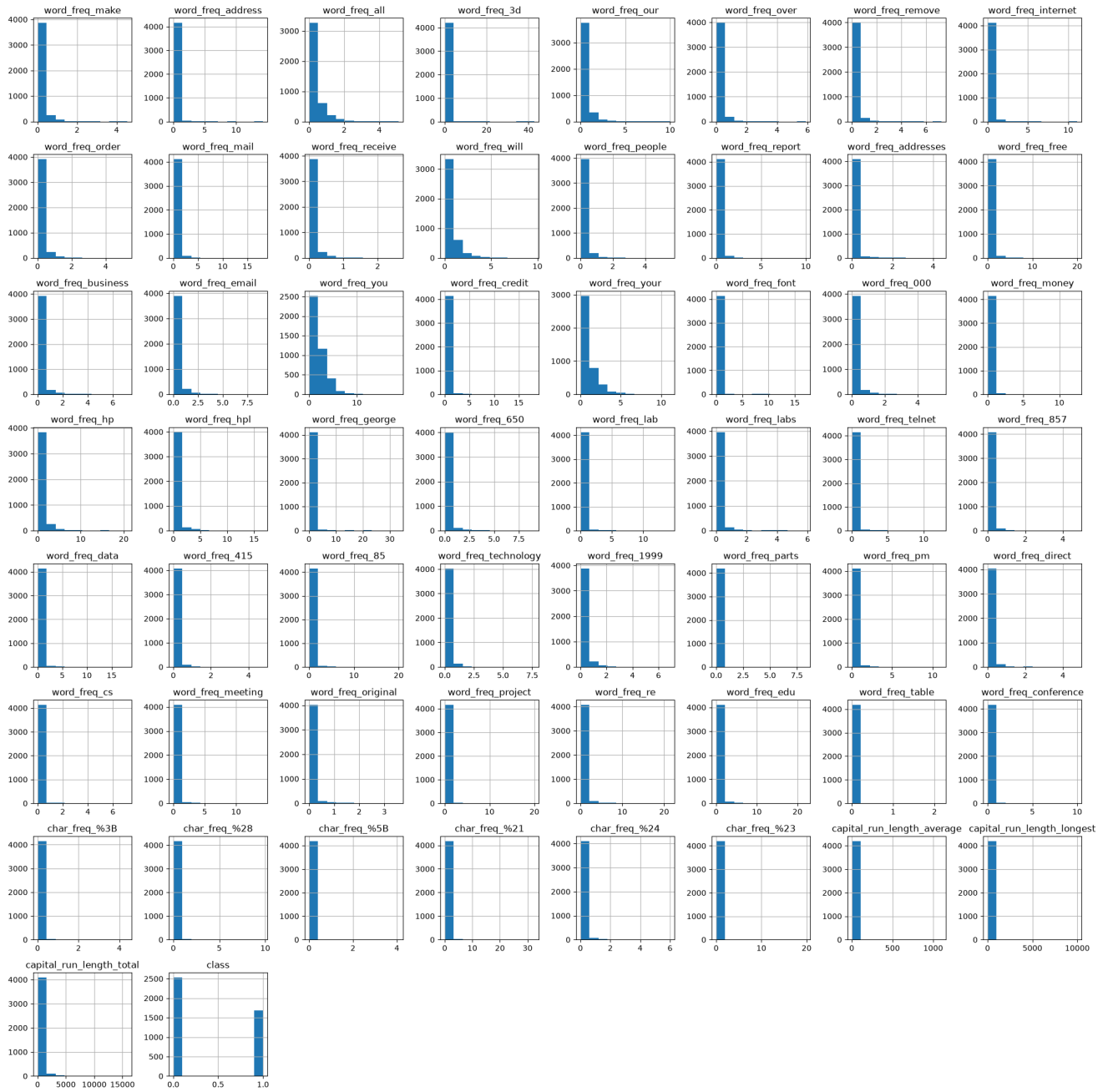


Figure 3: Distribution of Features

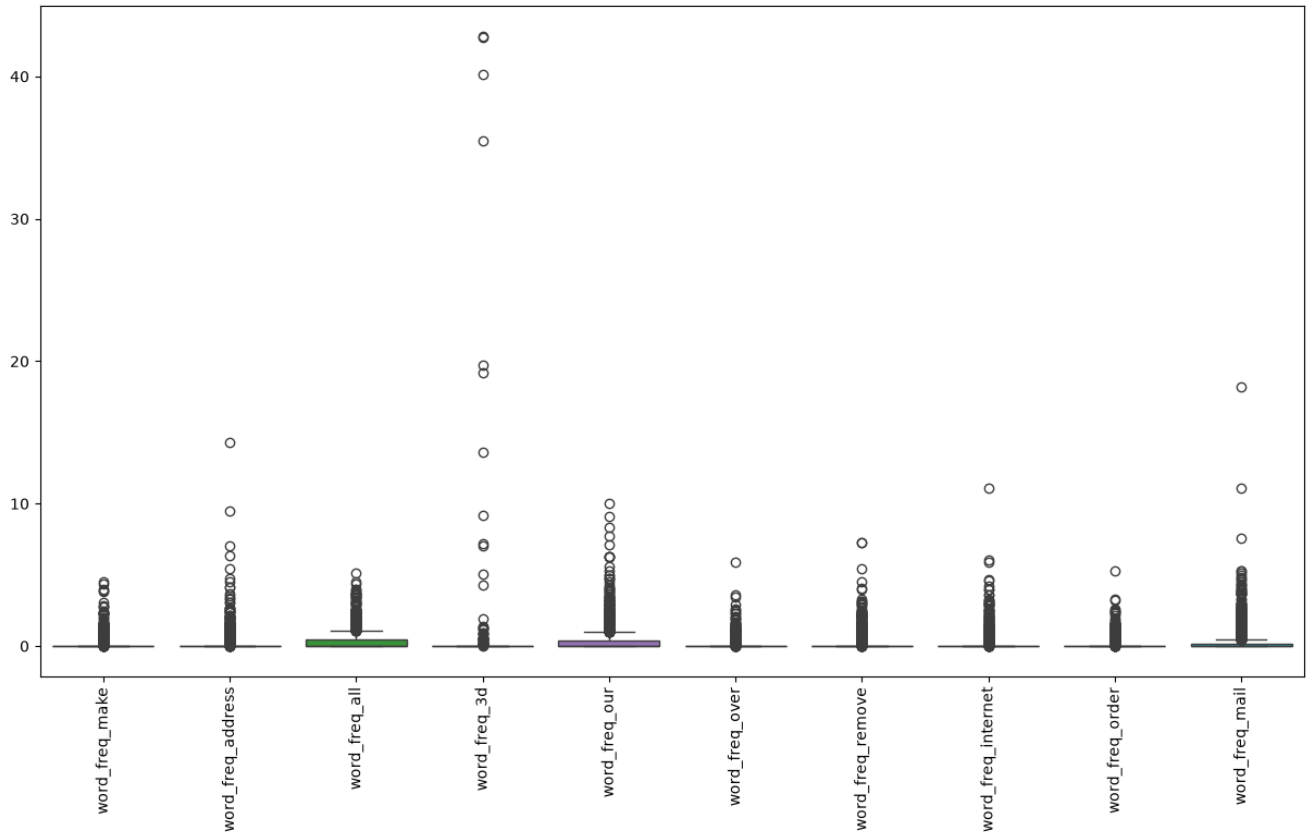


Figure 4: Box Plot of Selected Features

Inference:

- Most features are positively skewed.
- Several attributes contain outliers, which are common in email frequency data.
- Feature scaling is beneficial because the attributes have different ranges.
- The distributions justify the preprocessing performed before model training.

5 Data Preprocessing

The SpamBase dataset was preprocessed before training the machine learning models. The preprocessing steps performed are described below.

Missing Value Handling

The dataset was checked for missing values using the `isnull()` function. No missing values were found in any of the 58 attributes; therefore, no imputation or data cleaning was required.

Encoding

The SpamBase dataset consists entirely of numerical features representing word frequencies, character frequencies, and capital letter statistics. Since all input features were already in numerical format, no categorical encoding techniques such as Label Encoding or One-Hot Encoding were required.

Feature Scaling

Different feature scaling techniques were applied based on the requirements of each classification algorithm.

- **StandardScaler** was applied before training the Gaussian Naive Bayes and K-Nearest Neighbors (KNN) classifiers.
- **MinMaxScaler** was used for the Multinomial Naive Bayes classifier to transform feature values into the range of 0 to 1.
- **Binarizer** was applied before training the Bernoulli Naive Bayes classifier to convert numerical feature values into binary values.

Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio. Stratified sampling was employed to preserve the class distribution in both the training and testing datasets.

Feature Engineering

No additional feature engineering techniques were applied. The original numerical features extracted from the SpamBase dataset were used directly for model training after the required preprocessing steps.

6 Machine Learning Algorithms

The experiment implements three variants of the Naive Bayes classifier and the K-Nearest Neighbors (KNN) classifier for email spam detection.

6.1 Gaussian Naive Bayes

Working Principle

Gaussian Naive Bayes assumes that all features follow a Gaussian (Normal) distribution and that every feature is conditionally independent given the class label.

Mathematical Formulation

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

where C represents the class and X represents the feature vector.

Advantages

- Simple and computationally efficient.
- Performs well on high-dimensional datasets.
- Requires relatively small training data.

Limitations

- Assumes feature independence.
- Assumes Gaussian distribution of features.
- Performance decreases if assumptions are violated.

6.2 Multinomial Naive Bayes

Working Principle

Multinomial Naive Bayes is designed for count or frequency data. It estimates the probability of each class using the frequency of features.

Mathematical Formulation

$$P(C|X) \propto P(C) \prod_{i=1}^n P(x_i|C)$$

Advantages

- Suitable for text classification.
- Fast training and prediction.
- Works well with word-frequency features.

Limitations

- Requires non-negative feature values.
- Assumes feature independence.
- Sensitive to feature distribution.

6.3 Bernoulli Naive Bayes

Working Principle

Bernoulli Naive Bayes works with binary features by considering whether a feature is present or absent.

Mathematical Formulation

$$P(C|X) \propto P(C) \prod_{i=1}^n P(x_i|C)$$

where each feature takes either 0 or 1.

Advantages

- Effective for binary features.
- Fast and memory efficient.
- Performs well for text classification.

Limitations

- Requires binary input.
- May lose information after binarization.
- Assumes feature independence.

6.4 K-Nearest Neighbors (KNN)

Working Principle

KNN is a non-parametric supervised learning algorithm that classifies a sample based on the majority class of its nearest neighbours.

Mathematical Formulation

Euclidean Distance:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Manhattan Distance:

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

Advantages

- Simple and easy to implement.
- No explicit training phase.
- Effective for non-linear decision boundaries.

Limitations

- Computationally expensive during prediction.
- Sensitive to feature scaling.
- Performance depends on the choice of k .

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	Latest installed version
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel Core Ultra 7 Processor, 32 GB RAM

8 Hyperparameter Selection

Parameter	Value
n_neighbors	9
Weights	distance
Metric	manhattan
Algorithm	auto
Cross Validation	5-fold
Best CV Accuracy	0.9139

Table 2: Best Parameters Obtained Using GridSearchCV

9 Results

Metric	Value
Accuracy	0.9276
Precision	0.9148
Recall	0.8631
F1-score	0.8882
ROC-AUC	0.9546

Table 3: Performance of the Best KNN Classifier

10 Result Visualization

10.1 Correlation Matrix

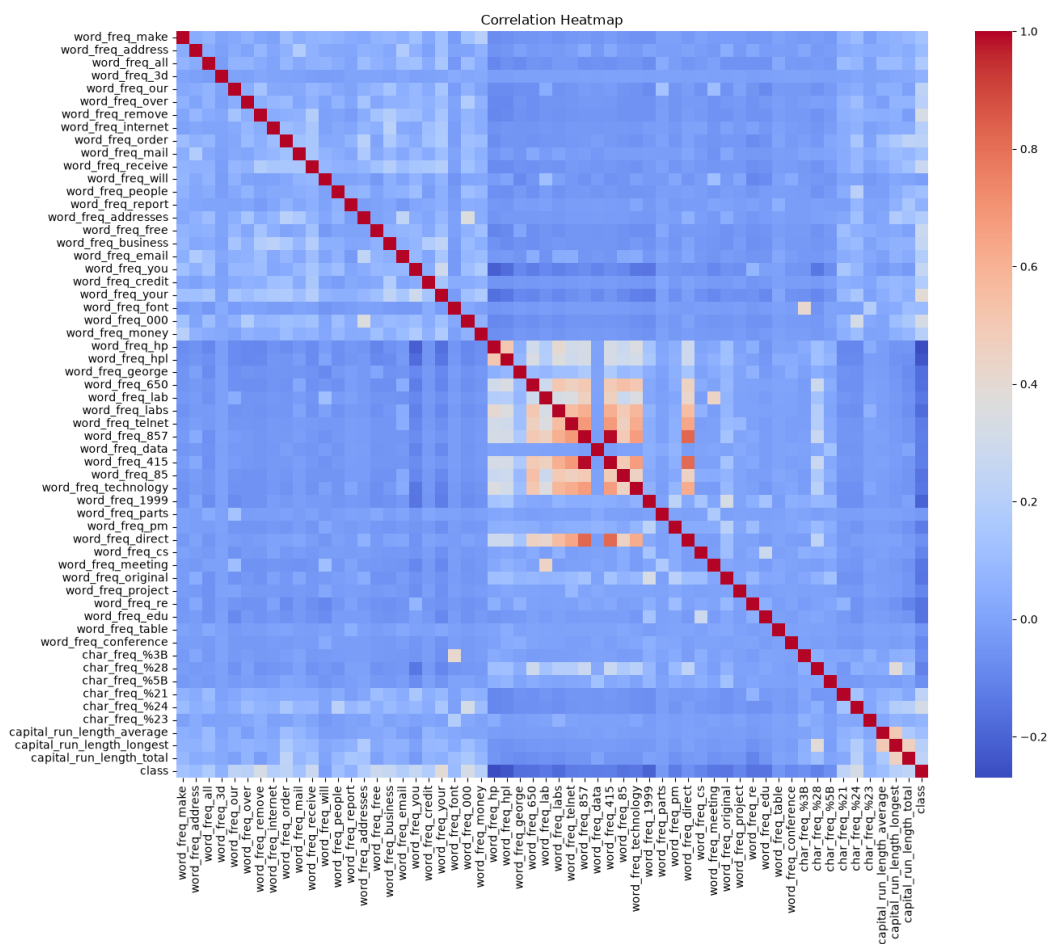


Figure 5: Correlation Matrix of SpamBase Features

Inference:

- The correlation matrix shows the relationship among numerical features.
- Most features exhibit weak to moderate correlation.
- No severe multicollinearity is observed among the input features.
- The heatmap helps understand feature relationships before model training.

10.2 Class Distribution

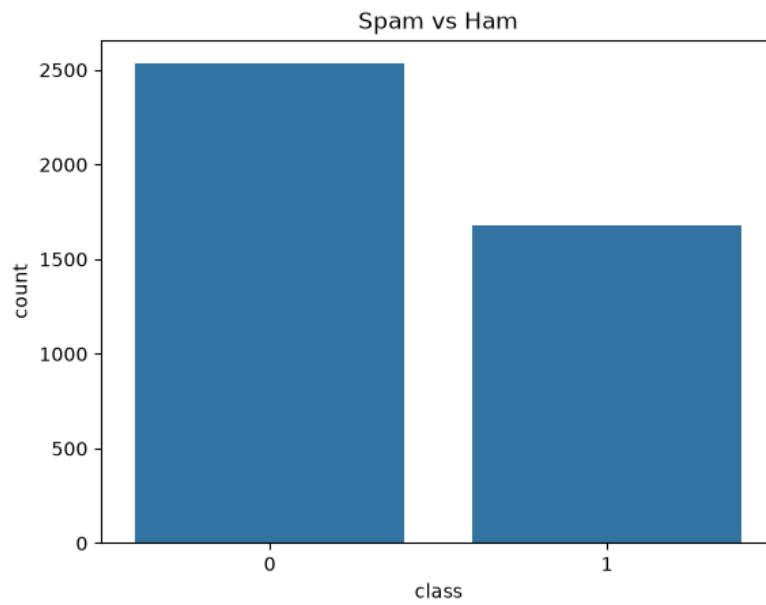


Figure 6: Distribution of Spam and Ham Emails

Inference:

- The dataset contains two classes: Spam and Ham.
- Both classes are well represented in the dataset.
- The class distribution is suitable for binary classification.
- Understanding the class balance helps evaluate classifier performance.

10.3 Confusion Matrix

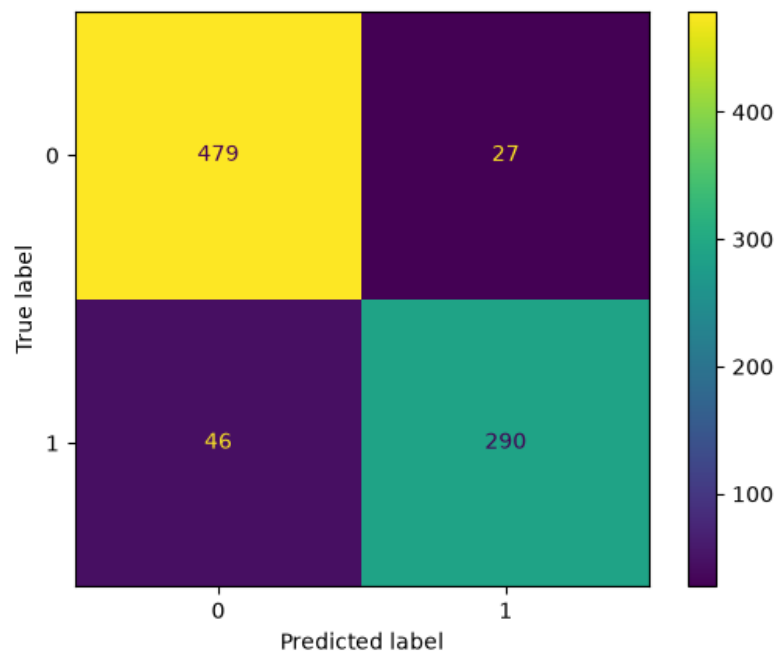


Figure 7: Confusion Matrix of the Best KNN Classifier

Inference:

- Most spam and ham emails are correctly classified.
- Only a small number of false positives and false negatives are observed.
- The confusion matrix indicates good classification performance.
- The classifier effectively distinguishes spam from legitimate emails.

10.4 ROC Curve

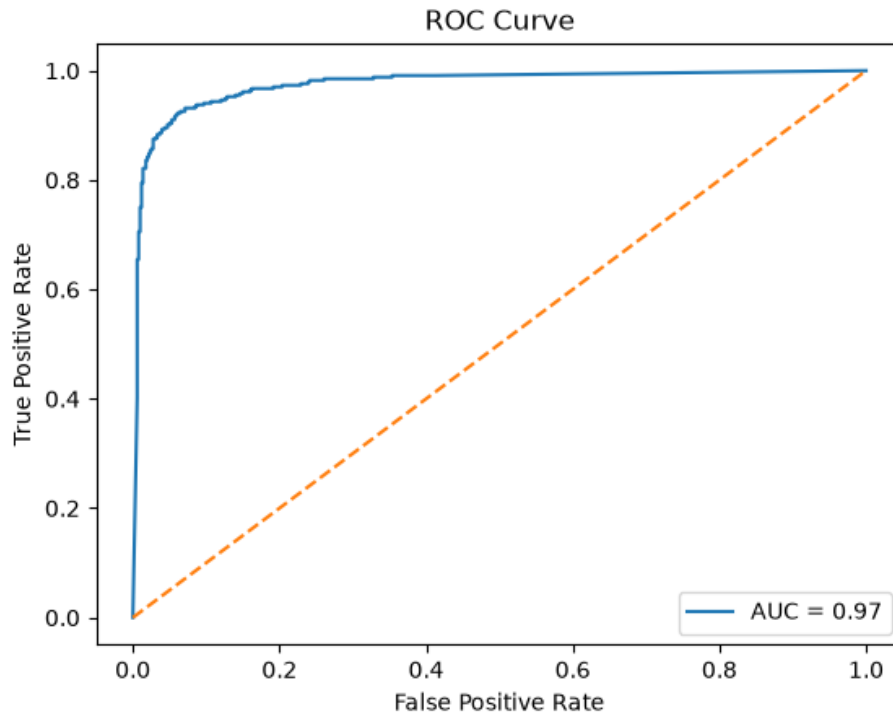


Figure 8: Receiver Operating Characteristic (ROC) Curve

Inference:

- The ROC curve lies close to the upper-left corner.
- The classifier achieves a high ROC-AUC score.
- A high ROC-AUC indicates excellent discrimination between spam and ham emails.
- The model performs well across different classification thresholds.

10.5 Precision–Recall Curve

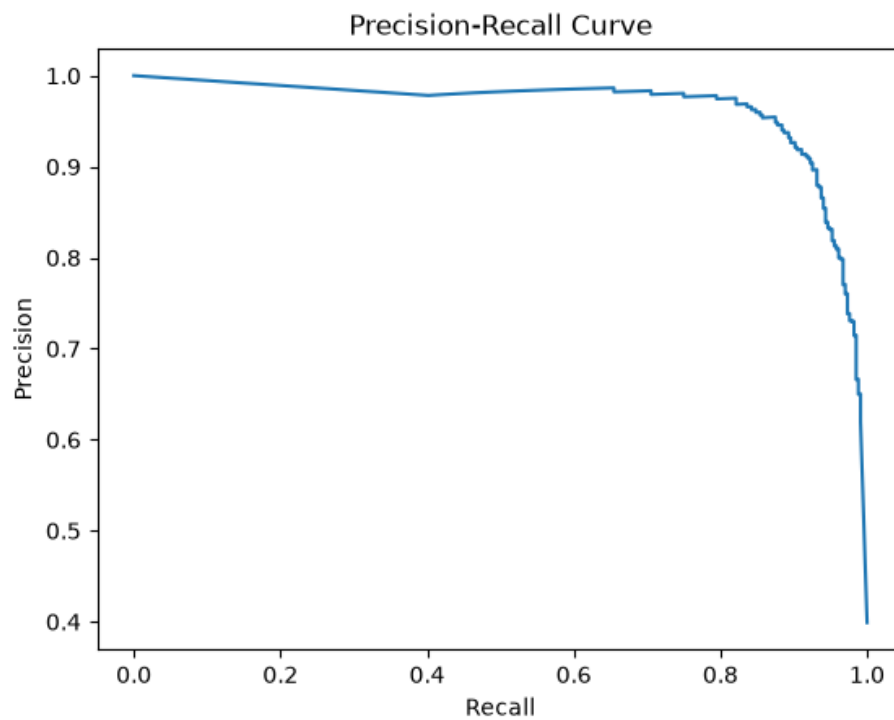


Figure 9: Precision–Recall Curve

Inference:

- The Precision–Recall curve demonstrates a good balance between precision and recall.
- Precision remains high across most recall values.
- The classifier maintains good performance on the positive (spam) class.
- This curve is especially useful for evaluating binary classification models.

10.6 Accuracy vs. K

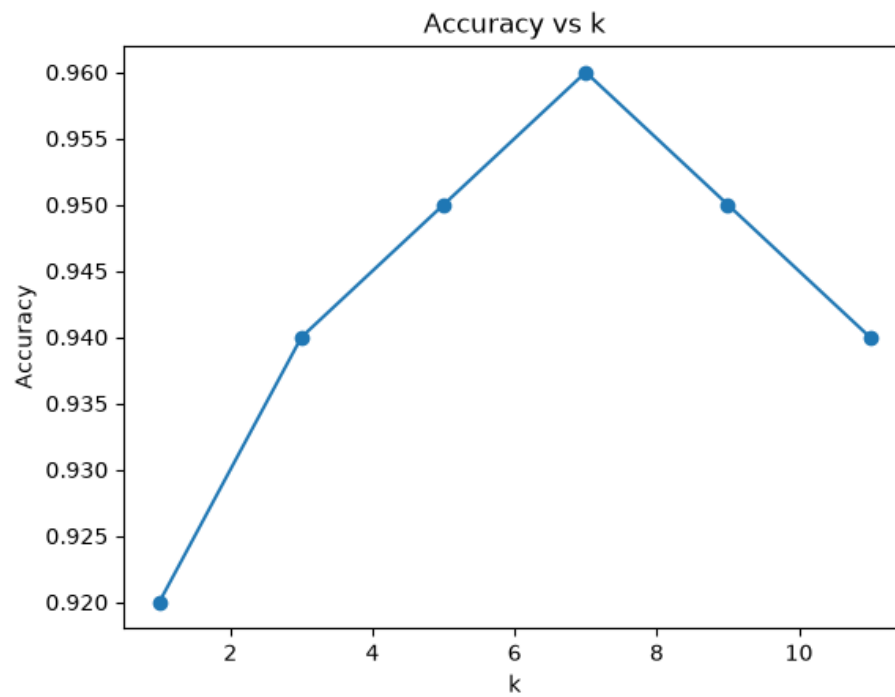


Figure 10: Accuracy for Different Values of K

Inference:

- Classification accuracy varies with different values of k .
- Selecting an appropriate value of k improves prediction performance.
- GridSearchCV identified the optimal value of k .
- This graph highlights the importance of hyperparameter tuning.

10.7 Classifier Comparison

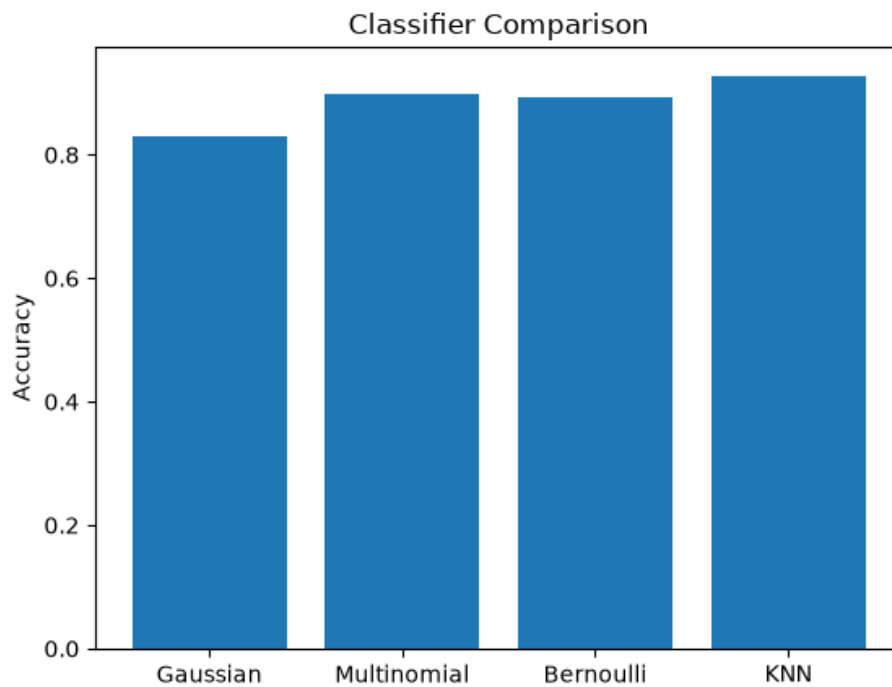


Figure 11: Comparison of Classification Algorithms

Inference:

- K-Nearest Neighbors (KNN) achieved the highest classification accuracy.
- Bernoulli Naive Bayes and Multinomial Naive Bayes also produced good results.
- Gaussian Naive Bayes showed comparatively lower performance.
- Overall, the tuned KNN classifier was the best-performing model in this experiment.

11 Discussion

Four classification algorithms were implemented and evaluated for spam email detection. Among the Naive Bayes classifiers, Bernoulli Naive Bayes achieved better performance than Gaussian and Multinomial Naive Bayes. Hyperparameter tuning using GridSearchCV improved the performance of the KNN classifier, resulting in the highest classification accuracy.

12 Conclusion

The SpamBase dataset was successfully used to classify emails as Spam or Ham using Naive Bayes and K-Nearest Neighbors classifiers. Appropriate preprocessing techniques, including feature scaling and train-test splitting, were performed before model training.

13 References

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4. Scikit-learn Documentation. <https://scikit-learn.org>